

Model	MCPMark					LOCA-bench					Toolathlon	MCP-Atlas
	 File.	 Notion	 Play.	 Post.	Avg.	8k	16k	32k	64k	Avg.		
Frontier Proprietary Models												
GPT-5.2	60.0	42.9	40.0	66.7	52.4	72.0	70.7	60.0	52.0	63.7	41.7 [†]	67.6
Claude Sonnet-4.5	32.5	25.0	27.0	50.0	33.6	58.7	60.0	36.0	26.7	45.3	38.9 [†]	43.8
Gemini-3 Pro	56.7	43.8	40.0	70.2	52.7	26.7	22.7	17.3	9.3	19.0	36.4 [†]	54.1
Open-Source Foundation Models												
Qwen3-8B	3.3	0.0	4.0	4.8	3.0	1.3	0.0	0.0	0.0	0.3	2.8	7.6
Qwen3-14B	3.3	0.0	0.0	9.5	3.2	1.3	0.0	0.0	0.0	0.3	1.9	12.4
Qwen3-235B-A22B	13.3	10.7	0.0	4.8	7.2	2.7	1.3	0.0	0.0	1.0	3.7	12.0
DeepSeek-V3.2	36.7	45.5	17.0	66.6	41.5	78.7	80.0	61.3	45.3	66.3	35.2 [†]	56.2
Open-Source Baselines												
TOUCAN-7B	0.0	0.0	0.0	4.8	1.2	0.0	0.0	0.0	0.0	0.0	0.0	7.0
Simulator-8B	3.3	0.0	4.0	4.8	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8
AWM-8B	3.3	0.0	4.0	4.8	3.0	0.0	0.0	0.0	0.0	0.0	1.9	8.0
EnvFactory-8B	10.0	0.0	4.0	9.5	5.9	<u>2.7</u>	<u>1.3</u>	0.0	0.0	<u>1.0</u>	1.9	11.8
EnvScaler-8B	10.0	0.0	4.0	9.5	5.9	1.3	0.0	0.0	0.0	0.3	<u>2.8</u>	<u>13.0</u>
Agent-World-8B	<u>13.3</u>	<u>3.6</u>	4.0	<u>19.1</u>	<u>10.0</u>	–	–	–	–	–	–	–
AWM-14B	3.3	0.0	4.0	9.5	4.2	<u>5.3</u>	<u>2.7</u>	0.0	0.0	<u>2.0</u>	<u>1.9</u>	<u>11.0</u>
Agent-World-14B	<u>16.6</u>	<u>3.6</u>	4.0	<u>38.1</u>	<u>15.6</u>	–	–	–	–	–	–	–
Our Method												
ProgAgent-8B	33.3	14.3	4.0	21.4	18.3	34.7	24.0	21.3	17.3	24.3	9.3	38.6
ProgAgent-14B	46.7	25.0	4.0	42.9	29.7	41.3	40.0	41.3	30.7	38.3	17.1	45.6